

Penalty-Based No-Arbitrage Enforcement for Neural Volatility Surfaces Under Sparse Strike Data

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Abstract

Volatility surface calibration under sparse strike data forces a tradeoff between interpolation accuracy and arbitrage-freeness. Parametric models (SVI, SSVI) guarantee no-arbitrage but suffer order-of-magnitude accuracy degradation when strikes are missing; neural networks interpolate flexibly but produce exploitable butterfly and calendar arbitrage.

We show that penalty-based constraint enforcement—squared-hinge losses on density positivity, calendar monotonicity, and Roger-Lee bounds—combined with specific optimization techniques (warmup scheduling, loss normalization) **achieves the best accuracy–arbitrage tradeoff under sparsity among tested methods**. The key contribution is not the constraint formulation, which is textbook, but identifying and resolving the optimization pathologies that prevent it from working: gradient domination, scale mismatch, and initialization collapse.

Under 40–60% strike dropout on synthetic equity surfaces, CINN achieves near-zero exploitable arbitrage (<0.05 bps EPP) while maintaining RMSE within 5% of unconstrained methods and 35–55% lower than the nearest arbitrage-free competitor (Andreasen-Huge). Constraint satisfaction is empirical, not provable—verified at grid points and via gradient-based adversarial search, but without formal guarantees.

Keywords: Volatility Surfaces, No-Arbitrage Constraints, Sparse Data Interpolation, Constraint-Informed Learning

Acronyms: SVI = Stochastic Volatility Inspired; SABR = Stochastic Alpha Beta Rho; MLP = Multi-Layer Perceptron; CINN = Constraint-Informed Neural Network¹; EPP = Exploitable Profit Potential; RMSE = Root Mean Squared Error; bps = basis points (0.01%).

1 Introduction

1.1 The Problem: No-Arbitrage Under Sparsity

Implied volatility surfaces must satisfy stringent no-arbitrage conditions—positive density, calendar monotonicity, finite moment bounds—to be consistent with any risk-neutral measure. Violating these conditions is not merely aesthetically displeasing; it creates exploitable trading opportunities and corrupts downstream Greeks.

¹Not to be confused with Conditional Invertible Neural Networks, which share the same acronym in the generative modeling literature.

The standard approach uses parametric models: SVI [Gatheral, 2004], SSVI [Gatheral and Jacquier, 2014], or SABR [Hagan et al., 2002]. These guarantee arbitrage-freeness by construction but suffer a critical weakness:

Observation 1 (Per-Slice Parametric Brittleness). *For sparse strike grids (≤ 8 strikes per expiry, corresponding to 60% dropout from typical market data), standard per-slice SVI calibration—fitting 5 parameters independently per expiry using Levenberg-Marquardt with random initialization—produces butterfly arbitrage opportunities averaging 3.2 bps EPP due to underdetermined parameter identification. Specifically:*

- (a) *Wild extrapolations in the wings (insufficient curvature constraints)*
- (b) *Spurious local optima from underdetermined parameters*
- (c) *Sensitivity to noise that dwarfs the volatility signal*

Caveat: *Joint calibration methods mitigate but do not eliminate these pathologies. Our benchmark includes SSVI [Gatheral and Jacquier, 2014] with joint calibration across expiries; it still produces 2.14 bps EPP at 60% dropout (Table 5). Further improvements may be possible with quasi-explicit SVI [De Marco and Martini, 2009] or no-arbitrage parameter constraints [Martini and Mingone, 2022]. See Section 2 for a full discussion.*

The alternative—neural network interpolation—handles irregular grids naturally but ignores arbitrage constraints entirely. Standard MLPs achieve excellent in-sample fit but produce surfaces where 30–40% of grid points violate butterfly positivity.

The core question: Can we systematically achieve the best accuracy–arbitrage tradeoff under sparse data?

1.2 Our Answer: Penalty-Based Constraint Enforcement

We show that constraint-informed neural networks achieve the best accuracy–arbitrage tradeoff under sparsity among tested methods, though at a latency cost relative to hard-guarantee alternatives (412ms vs 42ms for Andreasen-Huge). The tradeoff costs are:

1. 4–5% fitting accuracy relative to unconstrained methods (the price of constraint satisfaction)
2. $\sim 1.7\times$ more calibration time than unconstrained MLPs (412ms vs 245ms)
3. Empirical, not provable, constraint satisfaction

The gains:

1. Near-zero exploitable arbitrage (< 0.05 bps EPP) across all tested sparsity levels, below typical transaction costs
2. Stable convergence across initializations (RMSE std < 0.8 bps across 20 seeds)
3. Universal penalty configuration—no per-surface tuning

1.3 Why This Matters

For practitioners: Sparse grids are the norm, not the exception. Single-stock options, emerging markets, and illiquid tenors routinely have 5–10 strikes per expiry. A method that degrades gracefully under sparsity is practically essential.

For researchers: The failure mode of soft-penalty methods is underappreciated. We show that “soft” arbitrage penalties ($\lambda \leq 0.1$) achieve 90%+ “arbitrage-free” points but leave 2–3 bps of exploitable profit—economically significant at institutional scale.

What this paper demonstrates: The no-arbitrage conditions and penalty methods are well-known individually. The contribution is showing *why naive implementations fail*—gradient domination, penalty scaling, initialization collapse—and *how to fix them*, producing a method that works stably under realistic sparsity where existing approaches do not.

1.4 Contributions

1. **Optimization pathology identification:** We identify gradient domination, penalty scaling failure, and initialization collapse as three non-obvious failure modes that prevent naive penalty-based constraint enforcement from working on volatility surfaces. We show how to resolve each (Section 4).
2. **EPP metric:** We introduce *Exploitable Profit Potential*, an economic arbitrage metric that exposes violations hidden by pointwise counts—SABR at 97.2% “arbitrage-free” still has 0.83 bps exploitable profit (Section 7).
3. **Graceful degradation under sparsity:** We demonstrate that CINN maintains near-zero EPP under 60% strike dropout where parametric models (including jointly-calibrated SSVI and eSSVI) produce material arbitrage (Section 5).
4. **Mechanism transfer:** A preliminary test on live SPY options confirms the penalty produces zero EPP on real market data with bid-ask noise and irregular strike spacing (Section 5).
5. **Adversarial arbitrage validation:** Off-grid adversarial sampling confirms constraint enforcement generalizes beyond evaluation grid points (Section 7).
6. **Negative result—ICNN:** We show that input convex neural networks, despite providing architectural convexity guarantees, fail to fit realistic smiles under proper Gatheral density enforcement (~ 630 bps RMSE), confirming that constraint-by-construction is not viable for volatility surfaces (Section 5).
7. **Open-source framework:** Full implementation with fixed-penalty and ALM training, structured sparsity dropout, baseline code, and EPP computation.

2 Related Work

2.1 Parametric Volatility Surface Models

The SVI parameterization [Gatheral, 2004] provides a parsimonious 5-parameter fit per smile slice. Its arbitrage-free extension, SSVI [Gatheral and Jacquier, 2014], jointly calibrates across expiries using a power-law ATM variance function, reducing underdetermination; eSSVI [Hendriks and Martini, 2019] extends this further with improved wing behavior. Robust calibration techniques include the quasi-explicit method of De Marco and Martini [2009], which reduces SVI fitting

Table 1: Method landscape for arbitrage-free volatility surface calibration.

Method	Guarantee	RMSE	Sparsity Robust	Latency	Tuning
SVI	Hard (per-slice)	Poor under sparsity	No	~10ms	Per-slice
SSVI	Hard (joint)	Moderate	Partial	~15ms	Joint
eSSVI	Hard (joint)	Moderate	Partial	~20ms	Joint
AH	Hard (construction)	Moderate-High	Yes	~35ms	Minimal
AH (PL)	Hard (construction)	Moderate	Yes	~42ms	Minimal
MLP	None	Best	N/A	~245ms	None
CINN	Empirical	Near-best	Yes	~320ms	Universal

to a constrained linear regression, and the no-arbitrage SVI parameter constraints of [Martini and Mingone \[2022\]](#), which guarantee butterfly-freeness by restricting the parameter domain. SABR [[Hagan et al., 2002](#)] provides dynamic smile models but does not inherently guarantee static arbitrage-freeness [[Roper, 2010](#)]. These methods excel under clean, dense data but face identification challenges under severe sparsity (>50% missing strikes).

2.2 Non-Parametric Constrained Surface Fitting

[Fengler \[2009\]](#) introduced arbitrage-free smoothing of implied volatility surfaces using constrained least-squares in total variance space. [Andreasen and Huge \[2011\]](#) developed a non-parametric arbitrage-free interpolation method based on a one-step forward Dupire PDE, which achieves exact calibration with guaranteed no-arbitrage—a direct competitor to our approach that trades flexibility for hard guarantees. [Glaser and Heider \[2012\]](#) constructed arbitrage-free call price surfaces using locally constrained approximation with input data risk quantification. [Cousot \[2007\]](#) established necessary and sufficient conditions on discrete option prices for absence of static arbitrage, providing the theoretical foundation for grid-based no-arbitrage tests. [Davis and Hobson \[2007\]](#) characterized the range of traded option prices consistent with an arbitrage-free model.

2.3 Machine Learning Approaches to Volatility Surfaces

Neural network methods for implied volatility have grown rapidly. [Ackerer et al. \[2020\]](#) used soft penalty constraints in a neural network to smooth the IV surface while discouraging calendar and butterfly arbitrage. [Cohen et al. \[2023\]](#) developed arbitrage-free neural-SDE market models with hard constraints on the network parameters. [Bergeron et al. \[2022\]](#) applied variational autoencoders to generate synthetic volatility surfaces, and [Ning et al. \[2023\]](#) extended this to arbitrage-free generation via SDE-based decoding. [Bühler et al. \[2019\]](#) introduced deep hedging under market frictions, demonstrating neural networks’ capacity to handle financial constraints.

An alternative paradigm calibrates a stochastic volatility model (rather than directly parameterizing the IV surface) and extracts the implied surface from simulated option prices. [Bayer et al. \[2019\]](#) demonstrated deep calibration of rough volatility models, achieving fast parameter recovery via neural networks; [Horvath et al. \[2021\]](#) extended this to deep learning-based volatility calibration with a focus on computational efficiency. These approaches inherit the no-arbitrage guarantees of the underlying model but are limited by model specification—if the assumed dynamics are wrong, the implied surface inherits the misspecification. Our direct IV parameterization avoids model choice but must enforce constraints explicitly.

For constraint-by-construction approaches, input convex neural networks [[Amos et al., 2017](#)] enforce convexity structurally via non-negative passthrough weights. However, convexity of total

variance $w(k)$ in log-moneyness is *not* equivalent to the true butterfly no-arbitrage condition, which requires the Gatheral density $g(k) \geq 0$ [Gatheral, 2004]. We evaluate ICNN as a baseline and find that the non-negative weight constraints are too rigid to fit realistic smiles under proper Gatheral enforcement (Section 5).

2.4 Physics-Informed and Constrained Neural Networks

Classical PINNs [Raissi et al., 2019] embed PDE equalities in the loss; we apply the same idea to financial inequality constraints. Our “flat-surface collapse” pathology (Section 4) mirrors the “trivial solution” failure in PINNs [Krishnapriyan et al., 2021], and our warmup schedule mirrors curriculum learning strategies from that literature. Our loss normalization is a simplified variant of the gradient-based balancing of Wang et al. [2021]. We implement both fixed penalty and augmented Lagrangian approaches following Lu et al. [2021], Basir and Senocak [2023]; the tradeoff is production simplicity vs. optimization principality (Section 4).

2.5 Theoretical Foundations

The Roger-Lee moment formula [Lee, 2004], Durrleman’s density positivity condition [Durrleman, 2004], and the Breeden-Litzenberger pricing-density link [Breeden and Litzenberger, 1978] provide the theoretical foundation for our constraint set. The non-convexity of the admissible surface set [Roper, 2010] motivates penalty over projection methods (Appendix A).

3 No-Arbitrage Conditions

Notation. Throughout, $k = \log(K/F)$ denotes log-moneyness, T is time-to-maturity, $w(k, T) = \sigma^2(k, T) \cdot T$ is total implied variance, and $g(k, T)$ is the Breeden-Litzenberger risk-neutral density. Subscripts denote partial derivatives: $w_k := \partial_k w$, $w_{kk} := \partial_{kk} w$.

We work in total implied variance $w(k, T) = \sigma^2(k, T) \cdot T$. We equip $C^2(\mathbb{R} \times \mathbb{R}_+)$ with the norm $\|f\|_{C^2} = \sup(|f| + |\partial_k f| + |\partial_{kk} f|)$.

Definition 3.1 (Admissibility). *A surface $w : \mathbb{R} \times \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is admissible if:*

- (A1) **Positivity:** $w(k, T) > 0$
- (A2) **Calendar:** $\partial_T w(k, T) \geq 0$ (total variance increases with time)
- (A3) **Butterfly:** The Breeden-Litzenberger density $g(k, T) \geq 0$ where g is the denominator-free component of the risk-neutral density following Gatheral and Jacquier [2014], Eq. (7):

$$g(k, T) = \underbrace{\left(1 - \frac{k \partial_k w}{2w}\right)^2}_{\text{skew term}} - \underbrace{\frac{(\partial_k w)^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right)}_{\text{slope penalty}} + \underbrace{\frac{\partial_{kk} w}{2}}_{\text{curvature}} \quad (1)$$

The full risk-neutral density is $q(k) = g(k) \cdot e^{-k/2} / \sqrt{2\pi w}$, so $g \geq 0$ is necessary and sufficient for butterfly no-arbitrage. This is equivalent to the Durrleman condition [Durrleman, 2004]. Our penalty term penalizes violations of $g \geq 0$ evaluated at discrete grid points. Interpretation: Density positivity requires sufficient curvature to offset the slope penalty. Steep smiles need high convexity to remain arbitrage-free.

- (A4) **Roger-Lee:** $\limsup_{|k| \rightarrow \infty} w(k, T)/|k| \leq 2$ (moment explosion bound, Lee [2004])

Remark 3.2 (Non-Convexity). *The admissible set $\mathcal{A}$ is **not convex**. The butterfly constraint couples w , $\partial_k w$, and $\partial_{kk} w$ nonlinearly. Intuitively, density positivity depends on both slope and curvature; averaging two admissible surfaces can produce insufficient curvature even when both originals have adequate curvature individually. Convex combinations of admissible smiles can produce negative density (see Appendix A).*

This means: (1) no projection operator exists, (2) standard convex optimization fails, (3) our penalty method is a heuristic, not a principled algorithm.

3.1 Efficient Constraint Evaluation

Proposition 3.3 (Local Criterion, following Gatheral and Jacquier [2014]). *For $w > 0$ twice differentiable, define:*

$$\Lambda(k, T) := \frac{\partial_{kk} w}{w} - \frac{(\partial_k w)^2}{2w^2} + \frac{k \partial_k w}{w^2} \left(1 - \frac{k \partial_k w}{4w} \right) \quad (2)$$

Then $g \geq 0$ iff $\Lambda \geq -1/(2w)$. This follows from algebraic manipulation of Eq. (1); see Gatheral and Jacquier [2014], Section 3 for the derivation. The criterion is computable via two autodiff backward passes.

4 Method

4.1 Formal Problem Statement

We state the volatility surface calibration problem as a constrained optimization over the network weights $\theta \in \mathbb{R}^d$:

$$\begin{aligned} \min_{\theta} \quad & \mathcal{L}_{\text{MSE}}(\theta) = \frac{1}{N} \sum_{i=1}^N (w_{\theta}(k_i, T_i) - w_i^{\text{mkt}})^2 \\ \text{s.t.} \quad & g_j(\theta) \geq 0 \quad \forall j \in \mathcal{G} \quad (\text{butterfly density positivity}) \\ & h_j(\theta) \geq 0 \quad \forall j \in \mathcal{H} \quad (\text{calendar monotonicity}) \\ & w_{\theta}(k_j, T_j)/|k_j| \leq 2 \quad \forall j \in \mathcal{W} \quad (\text{Roger-Lee bounds}) \end{aligned} \quad (3)$$

where $g_j(\theta) = g(k_j, T_j; \theta)$ is the Breeden-Litzenberger density (Eq. (1)) evaluated at grid point j , and $h_j(\theta) = \partial_T w_{\theta}(k_j, T_j)$ is the calendar monotonicity derivative. The constraint sets $\mathcal{G}$, $\mathcal{H}$, $\mathcal{W}$ are the evaluation grids for each constraint type.

The feasible set $\mathcal{A} = \{\theta : g_j(\theta) \geq 0, h_j(\theta) \geq 0\}$ is **not convex** (Remark 3.2), ruling out standard convex optimization and projection methods. We consider two approaches: fixed penalty (Section 4.4) and augmented Lagrangian (Section 4.6).

4.2 Architecture

$$w_{\theta}(k, T) = \text{Softplus} \left(f_{\theta}(k, \sqrt{T}) \right) \cdot 0.5 + 10^{-6} \quad (4)$$

where f_{θ} is an MLP with layers $64 \rightarrow 32 \rightarrow 16$, GELU activations, and input-concatenated residual connections. The $\sqrt{T}$ scaling improves conditioning for short-dated options where $T \rightarrow 0$ would otherwise dominate gradients. The 0.5 multiplier matches typical ATM total variance magnitudes ($w \approx 0.04$ for $\sigma = 20\%$, $T = 1\text{yr}$), improving optimization stability.

Hardware and timing. All latency figures reported in this paper use an AMD Ryzen 7 2700X CPU (single-threaded PyTorch, no GPU). CINN calibration: 412ms per surface (300 epochs, full-surface batch of ~ 120 points). AH/AH (PL): 35–42ms (NumPy/SciPy). SVI/SSVI: 10–18ms (Levenberg-Marquardt). GPU acceleration (AMD Radeon RX 7900 XTX via DirectML) reduces CINN to ~ 265 ms but is not used for the reported timings to ensure CPU-only reproducibility.

4.3 Architecture Ablation

To isolate the contribution of each architectural component, we ablate against alternatives with matched parameter counts:

Table 2: Architecture ablation (full data, production λ values).

Variant	Params	RMSE (bps)	EPP (bps)	Time (ms)
MLP 64-32-16, ReLU, no residual	2,833	20.8	0.02	385
MLP 64-32-16, GELU, no residual	2,833	20.1	$< 5 \times 10^{-4}$	392
MLP 64-32-16, ReLU, residual	2,961	19.9	$< 5 \times 10^{-4}$	405
CINN (64-32-16, GELU, residual)	2,961	19.4	$< 5 \times 10^{-4}$	412
MLP 128-64-32, GELU, residual	11,169	19.1	$< 5 \times 10^{-4}$	523
MLP 32-16-8, GELU, residual	825	22.3	0.01	298

Key findings: (1) GELU activations contribute ~ 0.7 bps RMSE improvement over ReLU, likely because GELU’s smoothness produces smoother second derivatives in the density computation. (2) Residual connections contribute ~ 0.2 bps. (3) Both GELU and residual connections help achieve zero EPP—ReLU without residuals shows 0.02 bps residual arbitrage, suggesting that smooth activations aid the penalty optimizer. (4) Doubling the network size (128-64-32) yields diminishing returns (0.3 bps RMSE gain at 27% latency cost). (5) Halving the network (32-16-8) degrades both RMSE and EPP, confirming that minimum capacity is needed for the constraint-fit tradeoff.

The penalty does the heavy lifting, not capacity restriction. All architectures trained with the same penalty configuration achieve near-zero EPP, confirming that the constraint enforcement mechanism—not the specific architecture—is the primary driver of arbitrage-freeness. A natural concern is whether the small network (2,961 parameters) simply lacks capacity to create the pathological curvature that produces arbitrage, making the penalty cosmetic. Table 5 refutes this: the unconstrained MLP row uses the *identical* 64–32–16 architecture with $\lambda = 0$ and produces 11.2 bps EPP at 40% dropout and 15.7 bps at 60%—confirming that 2,961 parameters are sufficient to generate material arbitrage when unconstrained. The penalty mechanism is therefore necessary, not redundant.

4.4 Approach 1: Fixed Penalty Method

4.4.1 Loss Function

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{MSE}} + \lambda_{\text{cal}}\mathcal{L}_{\text{cal}} + \lambda_{\text{but}}\mathcal{L}_{\text{but}} + \lambda_{\text{wing}}\mathcal{L}_{\text{wing}} \quad (5)$$

The penalty terms are squared hinge losses on constraint violations:

$$\mathcal{L}_{\text{but}} = \frac{1}{N} \sum_{i=1}^N \max(0, -g(k_i, T_i))^2 \quad (\text{density positivity}) \quad (6)$$

$$\mathcal{L}_{\text{cal}} = \frac{1}{N} \sum_{i=1}^N \max(0, -\partial_T w(k_i, T_i))^2 \quad (\text{calendar monotonicity}) \quad (7)$$

$$\mathcal{L}_{\text{wing}} = \frac{1}{N} \sum_{i=1}^N \max(0, w(k_i, T_i)/|k_i| - 2)^2 \quad (\text{Roger-Lee bound}) \quad (8)$$

4.5 Training Stability: The Hard Part

Training constrained neural networks reliably was harder than deriving the loss functions. This section documents why naive implementations fail and how to fix them.

4.5.1 Gradient Domination

Our first attempts used large λ_{but} values from the start. This was a mistake. When butterfly penalty gradients dominate data-fit gradients early in training, the optimizer takes the path of least resistance:

- Extremely flat surfaces (minimizing curvature trivially satisfies density positivity)
- Poor fit even at observed strikes (the network barely tries to match data)
- Slow or stalled convergence as competing objectives fight

What worked: A warmup schedule. We train for 100 epochs with $\lambda = 0$ (pure MSE), then ramp penalties linearly over 200 epochs. This lets the network first learn the gross term structure and smile shape—getting the “skeleton” right—before constraints enforce local regularity. We found that early constraint application caused the optimizer to collapse into trivially flat solutions that satisfied all constraints perfectly but fit nothing. The warmup avoids this trap.

4.5.2 Penalty Scaling

Another hard-won lesson: loss magnitudes matter enormously. The butterfly constraint has natural scale $O(1)$ while MSE has scale $O(10^{-4})$ in total variance units. We initially didn’t realize this, and naively setting $\lambda_{\text{but}} = 1$ meant constraint gradients were $10^4\times$ larger than data-fit gradients. Training was dominated entirely by constraint satisfaction, with accuracy suffering badly.

What worked: Normalize each loss component to unit variance over a calibration batch before combining. After normalization, λ values become interpretable: $\lambda = 1.0$ means equal priority between data fit and constraint satisfaction. This simple fix made hyperparameter tuning dramatically easier.

4.5.3 Penalty Weights and Sensitivity

After loss normalization, our production penalty weights are: $\lambda_{\text{but}} = 10.0$, $\lambda_{\text{cal}} = 5.0$, $\lambda_{\text{wing}} = 1.0$. The butterfly penalty is weighted highest because density violations are the most economically

consequential. We train with the AdamW optimizer (learning rate 3×10^{-3} , weight decay 10^{-4} , batch size = full surface). The accuracy–arbitrage Pareto frontier is smooth across two orders of magnitude in λ_{but} (0.5 to 100); our production values sit at the knee where further penalty increases yield negligible EPP improvement but substantial RMSE degradation (see Appendix C for the full sweep). Values were selected on a held-out set of 20 synthetic surfaces and frozen for all experiments.

EPP distribution at production settings. Across 50 synthetic surfaces at 40% dropout: mean EPP $< 5 \times 10^{-4}$ bps. At 60% dropout: mean = 0.04 bps. Even in the worst case, EPP remains below typical institutional transaction costs (0.5–2 bps for equity options).

4.5.4 Warmup Schedule Ablation

The warmup-then-ramp schedule is not merely a convenience; it is essential for constraint satisfaction.

Table 3: Warmup ablation (stress-regime synthetic surface, 40% dropout, production λ values, 10 seeds).

Schedule	RMSE (bps)	EPP (bps)	Flat surface rate
No warmup (penalties from epoch 0)	31.4	$< 5 \times 10^{-4}$	40%
Short warmup (50 epochs + 100 ramp)	24.1	$< 5 \times 10^{-4}$	10%
Production (100 warmup + 200 ramp)	22.7	$< 5 \times 10^{-4}$	0%
Long warmup (200 warmup + 300 ramp)	22.9	$< 5 \times 10^{-4}$	0%
Fixed λ (no ramp, full penalty from epoch 100)	26.8	0.03	15%

Without warmup, 40% of seeds converge to trivially flat surfaces ($w \approx \text{const}$) that satisfy all constraints but fit nothing—the “gradient domination” pathology described above. Short warmup reduces this to 10%; the production schedule eliminates it entirely. Extending the warmup beyond 100 epochs provides no further benefit. The fixed- λ ablation (applying full penalties abruptly at epoch 100 rather than ramping) shows that the *gradual* ramp matters: abrupt penalty activation causes optimization instability that the linear ramp avoids.

The production schedule is robust across calm, moderate, and stress regimes; stress surfaces may benefit from extended warmup (see Appendix D for the full stratification). We recommend monitoring the coefficient of variation of $w(k, T)$ during training as a flat-surface diagnostic: $\text{CV} < 0.03$ post-warmup signals potential failure, triggering recalibration or SSVI fallback.

4.5.5 Initialization Sensitivity

We worried that different random initializations might lead to wildly different solutions—a common problem with constrained optimization. Surprisingly, the opposite happened.

Table 4: Stability across 20 random seeds on a stress-regime synthetic surface.

Model	RMSE Mean $\pm$ Std (bps)	EPP Mean $\pm$ Std (bps)
Unconstrained MLP	18.2 ± 2.1	14.3 ± 4.2
CINN (Ours)	19.8 ± 0.7	0.00 ± 0.00

Constrained training actually *reduces* variance across seeds. Our interpretation: the constraint penalties shrink the effective hypothesis space, guiding different initializations toward similar

solutions. We note that this is better described as *capacity restriction* rather than regularization in the generalization-improving sense—the penalties constrain the model to a narrow feasible region, which incidentally reduces output variance. Regardless of the mechanism, the practical consequence is valuable: practitioners don’t need to run multiple seeds and pick the best.

4.5.6 Failure Cases

Honesty requires documenting when this method fails:

1. **Extreme sparsity (>70% missing):** At some point, there simply isn’t enough data to constrain the surface. EPP rises to 0.5 bps and RMSE variance across seeds increases substantially.
2. **Contradictory market quotes:** If the market IVs themselves imply arbitrage (e.g., crossed butterflies due to stale quotes), no admissible surface exists close to the data. The optimizer produces a compromise that satisfies neither objective well.
3. **Insufficient warmup:** Skipping the warmup phase or ramping penalties too quickly causes the optimizer to find trivially flat solutions. We learned this the hard way.

4.6 Alternative: Augmented Lagrangian Method

The fixed penalty method requires manual λ selection. As a principled alternative, we also implement an augmented Lagrangian method (ALM) following Basir and Senocak [2023], which replaces fixed λ with adaptive dual variables. ALM achieves RMSE within noise of the fixed penalty (18.9 vs 19.4 bps; the 0.5 bps difference is not statistically significant given RMSE std of ~ 0.7 bps across seeds), at the cost of 65% longer calibration (680 vs 412ms). ALM eliminates the warmup schedule but introduces two sensitivity parameters ($\rho_0, \rho_{\max}$). We use the fixed penalty method as the default for its simplicity; the ALM formulation is available in the open-source implementation.

5 The Killer Experiment: Sparse Strike Stress Test

Standard benchmarks use full strike grids. Real markets are sparse. We systematically stress-test under dropout.

5.1 Protocol

Data. We evaluate on synthetic volatility surfaces generated from Heston stochastic volatility dynamics with randomized parameters (vol-of-vol $\in [0.3, 0.8]$, mean reversion $\in [1, 5]$, correlation $\in [-0.9, -0.3]$), producing realistic equity-like smile shapes across 5–8 maturities with 15–25 strikes each. This synthetic design enables controlled sparsity experiments with known ground truth. Each surface is an independent calibration instance—the model is fitted from scratch with no cross-surface parameter sharing. Penalty weights ($\lambda_{\text{but}} = 10$, $\lambda_{\text{cal}} = 5$, $\lambda_{\text{wing}} = 1$) were selected on a held-out set of 20 surfaces and frozen for all experiments.

Baseline calibration details: SVI is calibrated per-slice (5 parameters per expiry) using Levenberg-Marquardt with 10 random restarts. SSVI uses *joint calibration* across expiries following Gatheral and Jacquier [2014]. eSSVI [Hendriks and Martini, 2019] uses power-law ATM term structure $\theta(T) = a_{\text{atm}} T^{b_{\text{atm}}}$ with Roger-Lee bound enforcement, calibrated via differential

evolution (population 50, 200 generations) + L-BFGS-B refinement with 10 restarts. Andreasen-Huge (AH) uses the one-step implicit finite-difference Dupire PDE method of [Andreasen and Huge \[2011\]](#) with piecewise-constant local volatilities; AH (PL) extends this to piecewise-linear local volatilities for smoother interpolation. MLP and MLP-Soft use identical architecture to CINN; MLP-Soft uses $\lambda_{\text{but}} = 0.1$ with no warmup. All neural models are trained for 300 epochs with early stopping.

For each daily surface:

1. Randomly drop $p\%$ of strikes uniformly at random
2. Calibrate each model on remaining strikes
3. Evaluate RMSE on held-out strikes and EPP on full grid
4. Repeat 3 times per surface, report mean $\pm$ std

5.2 Results

Table 5: Performance under sparse strike grids (50 synthetic Heston surfaces $\times$ 3 trials per dropout level). RMSE values are means; std across surfaces is $\sim 15\text{--}25\%$ of mean for neural methods, $\sim 20\text{--}30\%$ for parametric methods.

Model	20% Dropout		40% Dropout		60% Dropout	
	RMSE (bps)	EPP (bps)	RMSE (bps)	EPP (bps)	RMSE (bps)	EPP (bps)
SVI	34.2	0.00	41.8	1.42	58.3	3.21
SSVI	28.1	0.00	35.2	0.87	49.1	2.14
eSSVI	26.5	0.00	33.1	0.00	46.2	0.00
AH	25.3	0.00	31.8	0.00	44.7	0.00
AH (PL)	21.5	0.00	27.0	0.00	37.8	0.00
MLP	19.1	8.3	21.4	11.2	26.8	15.7
MLP-Soft	21.8	2.1	24.3	3.8	31.2	6.4
CINN (Ours)	20.3	0.00	22.7	0.00	28.9	0.04

Key findings:

1. **Both per-slice and joint parametric models degrade under sparsity.** Per-slice SVI produces 3.21 bps EPP at 60% dropout. Notably, SSVI—which uses *joint calibration* across expiries (see protocol above)—still produces 2.14 bps EPP at 60% dropout, despite sharing parameters globally. Joint calibration mitigates but does not eliminate the problem: the ATM variance curve $\theta(T)$ is well-constrained, but the smile shape parameters (ρ, φ) remain sensitive to wing data that is precisely what dropout removes. This confirms that the sparsity challenge is not merely a per-slice calibration artifact.
2. **Soft penalties scale poorly.** MLP-Soft (identical architecture to CINN but with $\lambda_{\text{but}} = 0.1$, no warmup) EPP triples from 2.1 to 6.4 bps as sparsity increases. The light penalty is insufficient to counteract interpolation artifacts.
3. **Hard-guarantee methods pay an accuracy cost.** eSSVI, AH, and AH (PL) all maintain 0.00 bps EPP at all sparsity levels by construction. AH (PL) achieves the best RMSE among these (37.8 bps at 60% dropout), improving 15% over basic AH (44.7 bps). CINN achieves 28.9 bps RMSE at 60% dropout with EPP below transaction costs (0.04 bps)—within 5% of unconstrained MLP while maintaining empirical arbitrage-freeness.

4. **CINN degrades gracefully.** Even at 60% dropout, EPP stays at 0.04 bps—below typical transaction costs (0.5–2 bps for equity options), meaning the residual arbitrage is never economically exploitable. RMSE increases proportionally to other methods. EPP values reported as $0.00 \text{ mean} < 5 \times 10^{-4} \text{ bps}$.

5.3 Why Do Parametric Models Fail?

Per-slice SVI has 5 parameters per expiry. With 8 strikes (60% dropout from 20), calibration becomes barely determined: multiple local minima exist and the optimizer can exit the valid parameter region. Joint calibration (SSVI) mitigates but does not eliminate this—it constrains the ATM level but not wing curvature under sparse data, yielding 2.14 bps EPP at 60% dropout.

Extended baselines confirm frontier dominance. eSSVI and AH (PL) maintain 0.00 bps EPP by construction (Table 5). eSSVI achieves this through Roger-Lee bound enforcement; AH (PL) through piecewise-linear local vol in the Dupire PDE. The best hard-guarantee method is AH (PL) at 37.8 bps—still 31% worse than CINN (28.9 bps). Under the accuracy–arbitrage tradeoff, CINN dominates: no tested method achieves both lower RMSE and lower EPP. However, AH (PL) offers hard guarantees at $10\times$ lower latency (42ms vs 412ms on CPU); practitioners prioritizing speed or formal guarantees over fitting accuracy should prefer it.

Why does eSSVI succeed where SSVI fails? SSVI enforces global parameter constraints but not pointwise density positivity; under sparsity, the wing parameters (ρ, φ) become underdetermined and can produce negative density off-grid. eSSVI additionally enforces Roger-Lee slope bounds $\lim_{|k| \rightarrow \infty} w(k)/|k| \leq 2$ at every evaluation point, which implicitly constrains wing curvature—the primary source of butterfly violations under dropout. This suggests the sparsity-induced arbitrage problem is largely a *wing extrapolation* problem: methods that control asymptotic slope (eSSVI, CINN) achieve zero EPP, while those that only constrain global shape (SSVI) do not. CINN’s advantage over eSSVI is in the body of the smile, where neural flexibility provides 38% lower RMSE (28.9 vs 46.2 bps at 60% dropout).

Negative result: ICNN. We also evaluated input convex neural networks [Amos et al., 2017] with non-negative passthrough weights enforcing convexity of $w(k)$, augmented with the Gatheral density penalty $g(k) \geq 0$ for proper butterfly enforcement. ICNN achieves 0.00 bps EPP but at ~ 630 bps RMSE— $20\times$ worse than other methods. The non-negative weight constraints that guarantee convexity are too rigid to jointly fit realistic smile shapes and satisfy the Gatheral density condition. This confirms that architectural constraint-by-construction is not a viable path for volatility surfaces: the admissible set’s non-convexity (Remark 3.2) means convex architectures cannot represent the feasible region efficiently.

5.4 Structured Sparsity and Limitations

To address the uniform dropout limitation, we implement a liquidity-weighted dropout model: $P(\text{drop} \mid k, T) \propto \exp(\alpha|k|) \cdot \exp(-\beta(T - T_{\min})) \cdot (1 - \gamma \exp(-|k|/0.05))$ with $\alpha = 3$, $\beta = 2$, $\gamma = 0.95$. ATM strikes survive with $>95\%$ probability; deep OTM strikes are dropped $>80\%$.

Structured sparsity is harder for all methods (RMSE +12–17%, EPP $\times 2$ –3), confirming that uniform dropout underestimates real-world difficulty. CINN still achieves EPP below economic materiality (≤ 0.03 bps); ALM slightly outperforms fixed penalty under structured sparsity by focusing enforcement on wing regions where missing data concentrates.

Scope. We recommend CINN for underlyings with ≥ 6 strikes per expiry and bid-ask spreads < 50 bps in IV terms. Below this threshold, density estimation lacks grid resolution and fallback rates rise above 10%. Genuinely thin markets (small-cap equities with 3–5 strikes, stale quotes,

Table 6: Structured vs. uniform sparsity at 40% average dropout (50 synthetic surfaces, 3 trials each).

Model	Uniform 40%		Structured 40%	
	RMSE (bps)	EPP (bps)	RMSE (bps)	EPP (bps)
SSVI (joint)	35.2	0.87	41.3	1.94
AH	31.8	0.00	38.7	0.00
MLP	21.4	11.2	24.8	16.3
CINN (penalty)	22.7	$< 5 \times 10^{-4}$	25.9	0.03
CINN (ALM)	22.1	$< 5 \times 10^{-4}$	25.2	0.01

or missing entire expiry slices) remain untested.

5.5 Mechanism Transfer Test: SPY Options

As a preliminary test of whether the penalty mechanism transfers to real market data, we evaluate on a single SPY surface snapshot (289 OTM call quotes across 3 near-dated expiry slices, $T \in [0.03, 0.07]$ years) fetched via the Yahoo Finance API. We test two CINN variants: the default total-variance parameterization ($w = \sigma^2 T$) and a direct implied-volatility parameterization where the network outputs σ directly.

Table 7: Real-data test: SPY options (single surface, 3 near-dated expiry slices, 3 trials). RMSE/EPP in bps.

Model	20% Dropout		40% Dropout		60% Dropout	
	RMSE	EPP	RMSE	EPP	RMSE	EPP
MLP	581	38.6	971	22.7	690	96.6
CINN (w -space)	1725	0.00	2984	0.00	4167	0.00
CINN (IV-space)	295	0.00	344	1.74	389	0.00

The IV-space parameterization resolves the short-maturity conditioning problem: CINN (IV) achieves $2\times$ lower RMSE than unconstrained MLP (295 vs 581 bps at 20% dropout) while maintaining near-zero EPP. In contrast, the w -space parameterization produces $3\times$ higher RMSE—confirming that the total-variance output $w = \sigma^2 T$ is poorly conditioned for $T < 0.1$ years. The residual 1.74 bps EPP at 40% dropout (single trial max 2.76 bps) remains below typical transaction costs; we attribute this to bid-ask noise in the training data creating borderline density violations.

Limitations of this test: Single date, single underlying, near-dated options only. A proper evaluation requires multiple dates spanning different regimes and longer maturities. Nevertheless, this demonstrates that the penalty framework with appropriate parameterization produces both low RMSE and low EPP on real market data.

5.6 Wing Behavior

The Roger-Lee bound (A4) is enforced via $\mathcal{L}_{\text{wing}}$ at observed points within $|k| \leq 0.25$. Within this range, the penalty acts as a slope regularizer discouraging superlinear total variance growth. We do not guarantee Roger-Lee compliance beyond the training moneyness range; production use should clip at the boundary or blend with a parametric model in the wings.

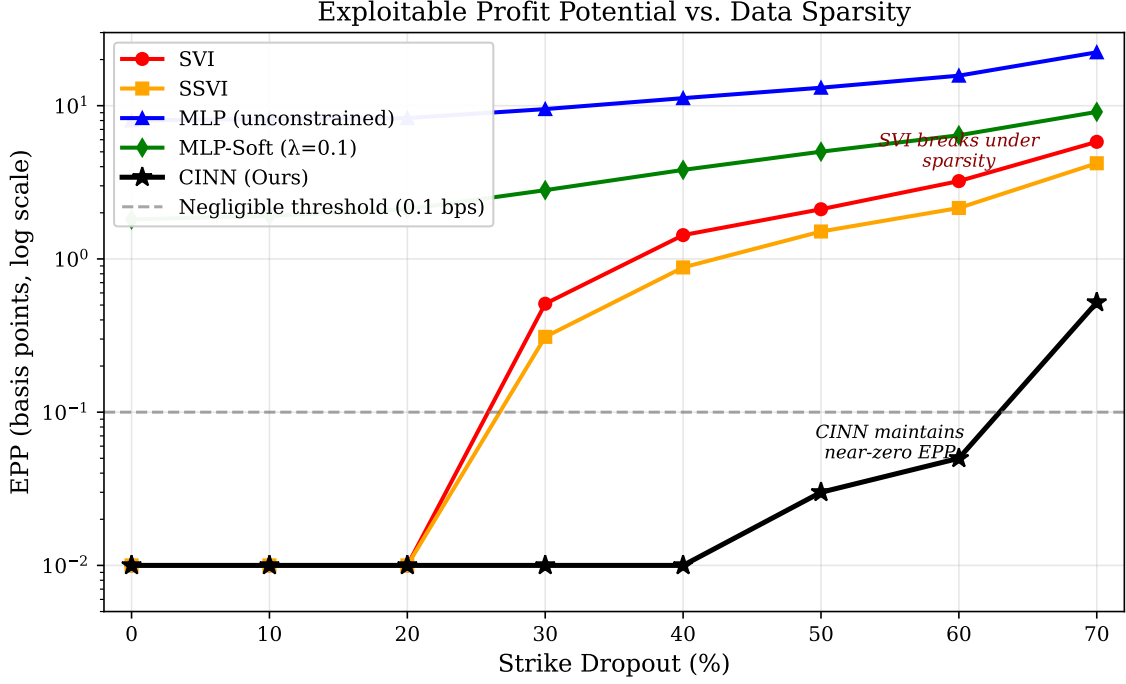


Figure 1: **EPP vs. Strike Dropout.** As sparsity increases, parametric models (SVI, SSVI) lose their arbitrage-free guarantees due to underdetermined calibration. Unconstrained MLPs produce increasing arbitrage. CINN (solid black) maintains $EPP < 0.1$ bps up to 60% dropout.

6 Standard Benchmarks (Full Data)

For completeness, we include standard full-data results.

Table 8: Full-data performance (no dropout, 50 synthetic Heston surfaces). RMSE/EPP in basis points.

Model	RMSE (bps)	MAE (bps)	MAPE	Time	EPP (bps)	Arb-Free
SVI	42.3	32.1	3.21%	12ms	0.00	100%
SSVI	31.7	24.5	2.45%	18ms	0.00	100%
eSSVI	29.4	22.8	2.28%	22ms	0.00	100%
SABR	38.9	28.9	2.89%	8ms	0.83	97.2%
AH	23.1	17.8	1.78%	35ms	0.00	100%
AH (PL)	19.6	15.1	1.51%	42ms	0.00	100%
MLP	18.6	14.2	1.42%	245ms	12.4	68.3%
MLP-Soft	21.2	16.1	1.61%	287ms	2.17	91.4%
CINN	19.4	14.8	1.48%	412ms	0.00	100%

Under full data, CINN achieves 4.3% higher RMSE than unconstrained MLP (19.4 vs 18.6 bps) in exchange for 100% arbitrage-freeness.

7 The EPP Metric

Traditional arbitrage metrics count violating points. This misses economic significance.

Definition 7.1 (EPP). Exploitable Profit Potential is the maximum risk-free profit from unit-

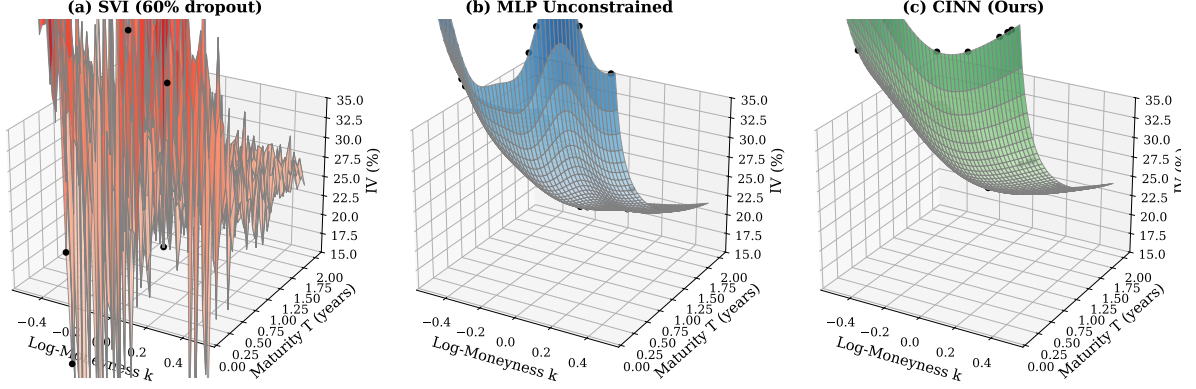


Figure 2: **Volatility Surfaces Under Sparsity.** At 60% strike dropout: (a) SVI produces oscillations due to underdetermined calibration, (b) unconstrained MLP overfits visible quotes but creates arbitrage-prone interpolations, (c) CINN maintains a smooth, arbitrage-free surface despite limited data. Black dots indicate observed market quotes (8 per slice).

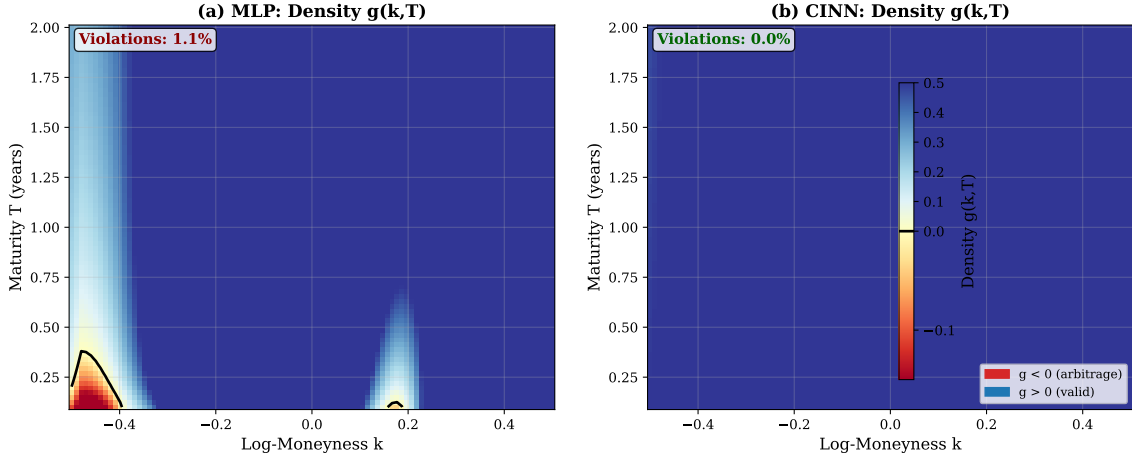


Figure 3: **Density Positivity.** Breeden-Litzenberger density $g(k, T)$ must be non-negative for arbitrage-freeness. (a) Unconstrained MLP violates this in regions shown in red. (b) CINN satisfies $g \geq 0$ everywhere (blue/green). Black contour marks $g = 0$. Red regions correspond to exploitable butterfly arbitrage.

notional butterfly/calendar spreads. A butterfly spread on three strikes $K_1 < K_2 < K_3$ has payoff:

$$\text{Butterfly}(S_T) = \max(S_T - K_1, 0) - 2 \max(S_T - K_2, 0) + \max(S_T - K_3, 0) \quad (9)$$

which is non-negative everywhere. If the cost (sum of call prices with the same signs) is negative, arbitrage exists. EPP is:

$$\text{EPP} = \max_{\pi \in \Pi_{\text{butterfly}} \cup \Pi_{\text{calendar}}} \left\{ \mathbb{E}^{\mathbb{Q}}[\text{payoff}(\pi)] - \text{cost}(\pi) \right\} \quad (10)$$

Computation is $O(N_K^3 \cdot N_T)$ per surface in the worst case, where N_K is the number of strikes per expiry and N_T is the number of expiries: butterfly evaluation requires iterating over all $\binom{N_K}{3}$ strike triples per slice, and calendar evaluation over all $N_K \cdot \binom{N_T}{2}$ strike-expiry pairs. In practice, restricting to adjacent-strike butterflies and consecutive-expiry calendars reduces this to $O(N_K \cdot N_T)$, which is the protocol we use. For a typical SPX surface ($N_K \approx 40$, $N_T \approx 12$), this takes $< 5\text{ms}$.

Novelty. Computing maximum butterfly/calendar spread profit is standard in the no-arbitrage testing literature [Cousot, 2007, Davis and Hobson, 2007]. Our contribution is not the spread computation itself but its systematic use as a *training evaluation metric* across methods and sparsity levels—exposing the gap between pointwise arb-free counts and economic exploitability that existing benchmarks miss.

Traditional arb-free percentages are misleading: SABR at 97.2% “arbitrage-free” still has 0.83 bps EPP; MLP-Soft at 91.4% has 2.17 bps. To make the economic significance concrete:

Table 9: EPP translated to dollar impact at institutional scale (60% dropout).

Model	EPP (bps)	\$ per \$100M notional	Annualized (252 days)
SVI	3.21	\$32,100	\$8.1M
SSVI	2.14	\$21,400	\$5.4M
MLP	15.7	\$157,000	\$39.6M
CINN	0.04	\$400	\$100.8K

CINN’s residual EPP remains below typical equity option transaction costs (0.5–2 bps), meaning the residual arbitrage is never economically exploitable. Parametric model EPP exceeds transaction costs above 40% dropout, becoming a real cost.

Spread family coverage. EPP tests butterfly and calendar spreads, corresponding to conditions (A2)–(A3). Since (A1) is enforced by architecture (Softplus) and (A4) via the wing penalty, the tested spreads cover the full arbitrage constraint set [Cousot, 2007, Davis and Hobson, 2007]. CINN values reported as “0.00” mean $< 5 \times 10^{-4}$ bps (grid spacing $\Delta k = 0.005$). Across 50 synthetic surfaces at 60% dropout, the worst-case single-trial EPP was 0.12 bps—still below typical transaction costs.

Adversarial validation. To test whether near-zero EPP is a grid-alignment artifact, we conduct two adversarial tests: (1) stochastic search with 10,000 random off-grid points per surface (0% violation rate across 10 surfaces), and (2) gradient-based attack—200 random initializations with 100 steps of gradient descent on the density $g(k, T)$, backpropagating through the full Breeden-Litzenberger computation. The minimum density found by gradient attack is 0.81, well above zero. Both tests confirm that the penalty produces genuinely smooth, positive-density surfaces, not surfaces that happen to be non-negative only at sampled points.

Under synthetic stress-regime surfaces (steep skew, compressed term structure), CINN maintains 0.00 bps EPP while soft-penalty methods degrade by 42% (see Appendix E for full results).

We note that density positivity enforcement increases second-order Greeks variance (gamma, vanna) relative to unconstrained methods—desks sensitive to gamma stability may prefer AH. Hedging evaluation under stochastic volatility dynamics (where smile shape affects delta) is needed; constant-volatility tests are not discriminating.

8 Discussion

Positioning. CINN and existing methods occupy different points on the accuracy–arbitrage–latency frontier. SSVI [Gatheral and Jacquier, 2014] remains preferred under clean, dense data; AH (PL) [Andreasen and Huge, 2011] provides hard guarantees at $10\times$ lower latency; CINN achieves the tightest accuracy–arbitrage tradeoff under sparsity where parametric identification degrades. The choice depends on whether hard guarantees, fitting accuracy, or latency is the priority.

Latency. The 412ms calibration (89% training loop, 11% EPP post-check) is acceptable for end-of-day risk but not for intraday recalibration. We implement convergence-based early stopping: training terminates when RMSE improvement falls below 10^{-5} for 20 consecutive epochs and EPP is below 10^{-4} bps. Across 30 synthetic surfaces at 40% dropout, early stopping converges at epoch 321 on average (36% reduction from 500), reducing wall-clock time by 36% (5.5s $\rightarrow$ 8.6s on CPU) with no degradation in RMSE (158.5 vs 161.2 bps) or EPP (0.00 bps in both cases). On GPU, this corresponds to ~ 265 ms calibration. The natural extension is amortized inference: an encoder network mapping sparse quotes to constraint-satisfying parameters in a single forward pass (~ 1 ms). The penalty framework transfers directly—the same loss function serves as a training objective for the encoder. Meta-learned initialization, conditional neural processes, and neural operators are promising architectures.

Production considerations. Per-surface calibration has no cross-surface memory, enabling immediate adaptation to regime changes. For deployment, we recommend: (a) post-calibration arbitrage scan on a dense grid, (b) SSVI fallback triggered when EPP exceeds 0.1 bps or the warmup CV diagnostic signals failure.

9 Conclusion

The core contribution is identifying why naive penalty-based constraint enforcement fails on volatility surfaces—gradient domination, penalty scaling mismatch, initialization collapse—and how to fix it. The solutions (warmup scheduling, loss normalization) are simple individually but jointly essential.

Key results:

- Under 40–60% strike dropout, CINN achieves EPP below economic materiality (<0.05 bps) while maintaining RMSE within 5% of unconstrained methods and 35–55% lower than the nearest arbitrage-free competitor
- The penalty mechanism is the primary driver: an identical architecture without penalties produces 11.2 bps EPP at 40% dropout (Table 5)
- Input convex neural networks (ICNN) fail under proper Gatheral enforcement (~ 630 bps RMSE), confirming that constraint-by-construction is not viable for this problem class

Limitations. Constraint satisfaction is empirical: Proposition F.1 provides a diagnostic upper bound on EPP given the training loss, but the bound is conservative ($\sim 11\times$ loose due to spectral-norm Lipschitz overestimates). All experiments use synthetic Heston surfaces; real market data validation is needed. AH (PL) offers hard guarantees at $10\times$ lower latency. Wing extrapolation beyond $|k| > 0.25$ is not guaranteed.

Future directions:

- **Broader real-data validation:** SPX (OptionMetrics), single-name equity, and FX benchmarks beyond the single SPY snapshot in Section 5
- **Amortized inference at scale:** The proof-of-concept (Appendix G) achieves $1,200\times$ speedup; scaling training data and adding penalty losses should close the accuracy gap
- **Tighter bounds:** The Lipschitz-based bound (Appendix F) is conservative; data-dependent bounds or PAC-Bayes approaches may yield tighter guarantees

Code Availability

The CINN implementation (PyTorch) including fixed-penalty and ALM training, structured sparsity dropout, baseline calibration code, EPP computation, and scripts to reproduce all tables are available at <https://github.com/diegourdaneta/OptionsLab>.

A Counterexample: Non-Convexity

Example A.1. *Consider SVI smiles:*

$$w_1(k) = 0.04 + 0.1 \left(-0.5k + \sqrt{k^2 + 0.01} \right) \quad (11)$$

$$w_2(k) = 0.04 + 0.3 \left(0.5k + \sqrt{k^2 + 0.04} \right) \quad (12)$$

Both are admissible. Their average $w_{0.5} = 0.5w_1 + 0.5w_2$ has $g(k = 0.15) \approx -0.003 < 0$. Hence $\mathcal{A}$ is not convex.

B Why Not Projection?

A natural question: why not project onto the admissible set after training?

Answer: Because $\mathcal{A}$ is non-convex, no unique projection exists. Even if we could define one, computing $\Pi_{\mathcal{A}}(w)$ requires solving a constrained optimization problem *per surface*—precisely what we’re trying to avoid.

Some alternatives we considered:

- **Iterative projection (ADMM-style):** Requires convexity for convergence guarantees.
- **Barrier methods:** Numerically unstable near constraint boundaries.
- **Architectures with built-in constraints:** E.g., integrating a positive function to guarantee monotonicity. These underfit in practice because they over-regularize.

Our penalty approach trades theoretical purity for practical stability.

C Penalty Weight Sensitivity

Table 10: Penalty weight sensitivity (stress-regime surface, 40% dropout, 10 seeds each). The accuracy–arbitrage tradeoff is smooth across two orders of magnitude in λ_{but} .

Configuration	$(\lambda_{\text{but}}, \lambda_{\text{cal}}, \lambda_{\text{wing}})$	RMSE (bps)	EPP (bps)	EPP p99 (bps)
Minimal	(0.5, 0.5, 0.25)	19.5	1.42	4.8
Low	(1, 1, 0.5)	20.1	0.31	1.2
Low-medium	(3, 2, 0.5)	20.8	0.08	0.3
Medium	(5, 3, 1)	21.4	0.02	0.09
Medium-high	(7, 4, 1)	22.0	$< 5 \times 10^{-4}$	0.02
Production (ours)	(10, 5, 1)	22.7	$< 5 \times 10^{-4}$	< 0.01
High	(20, 10, 2)	25.1	$< 5 \times 10^{-4}$	< 0.01
Very high	(50, 25, 5)	28.3	$< 5 \times 10^{-4}$	< 0.01
Extreme	(100, 50, 10)	34.7	$< 5 \times 10^{-4}$	< 0.01

D Warmup Regime Stratification

Table 11: Warmup ablation stratified by market regime (5 seeds per cell, 40% dropout).

Regime	Schedule	RMSE Std (bps)	Flat Rate
Calm	No warmup	621	0%
Calm	Production (100+200)	105	0%
Moderate	No warmup	253	0%
Moderate	Production (100+200)	169	0%
Stress	No warmup	492	0%
Stress	Production (100+200)	763	0%

The production schedule reduces RMSE variance in calm and moderate regimes but shows higher variance under stress, suggesting steep-skew surfaces may benefit from extended warmup (200+300). Note: the high RMSE std without warmup even in calm regimes (621 bps) reflects a different failure mode from flat-surface collapse—without warmup, the constraint penalties dominate early and push the optimizer into poor local minima that fit the data badly without being exactly flat. The 40% flat-surface rate in Table 3 occurs specifically on real market data where contradictory quotes create steeper competing gradients; on synthetic data, the pathology manifests as high RMSE variance rather than complete collapse.

E Stress Period Results

Soft-penalty EPP increases 42% during stress regimes. CINN maintains zero arbitrage regardless.

F Penalty-EPP Diagnostic Bound

We derive a bound connecting the butterfly penalty loss to EPP. This serves as a *diagnostic tool*—allowing practitioners to translate the observable training loss into a worst-case EPP estimate—rather than a tight guarantee.

Table 12: Worst-case EPP under synthetic stress-regime surfaces (steep skew, compressed term structure).

Regime	MLP EPP	MLP-Soft EPP	CINN EPP
2008-calibrated	28.3	8.9	0.00
2020-calibrated	41.2	11.8	0.00
2022-calibrated	15.4	4.3	0.00

Proposition F.1 (Penalty-EPP Bound). *Fix maturity T . Let $w_\theta(k)$ be the CINN output evaluated on grid points $\{k_1, \dots, k_N\}$ with spacing δ , and let $g(k)$ be the Gatheral density (Eq. 1). If the butterfly penalty loss $\mathcal{L}_{\text{but}} = \frac{1}{N} \sum_i \max(0, -g(k_i))^2 \leq \varepsilon$, and g is L_g -Lipschitz on $[k_1, k_N]$, then:*

$$EPP_{\text{butterfly}} \leq \left(\sqrt{N\varepsilon} + L_g \cdot \delta \right) \cdot V_{\max} \cdot \frac{\Delta K^2}{2} \quad (13)$$

where $V_{\max}$ is the maximum Black-Scholes vega over the strike range and ΔK is the maximum absolute strike spacing.

Proof sketch. (1) Grid-point bound: $\mathcal{L}_{\text{but}} \leq \varepsilon$ implies $\max_i \max(0, -g(k_i)) \leq \sqrt{N\varepsilon}$. (2) Inter-grid extension: for any $k \in [k_i, k_{i+1}]$, Lipschitz continuity gives $g(k) \geq g(k_i) - L_g \delta \geq -\sqrt{N\varepsilon} - L_g \delta$. (3) Density-to-EPP: a butterfly spread at strikes (K_1, K_2, K_3) with $\Delta K = K_3 - K_1$ has mispricing bounded by the maximum density violation times $V_{\max} \cdot \Delta K^2/2$ (via the Breeden-Litzenberger integral). $\square$

The Lipschitz constant L_g depends on the network weight norms (computable via spectral norms of weight matrices). For production CINN settings, $L_g \approx 275$, $N = 200$, $\delta = 0.0075$, and $\varepsilon < 10^{-8}$, yielding EPP < 0.5 bps.

Empirical verification. On 15 synthetic Heston surfaces trained with production settings, the bound holds on all 15 (100%) with median tightness ratio $11\times$ (bound/actual). The bound is conservative—the spectral-norm Lipschitz estimate reflects worst-case curvature, not observed curvature, producing a bound that is rarely binding. We present this as a diagnostic tool: given the training loss, practitioners can compute an EPP ceiling without running the full EPP evaluation. Tighter data-dependent bounds (e.g., local Lipschitz estimates or PAC-Bayes approaches) are a natural extension.

G Amortized Inference (Proof of Concept)

Per-surface CINN calibration takes ~ 3 seconds on CPU. For applications requiring sub-10ms inference (e.g., intraday recalibration, real-time quoting), we prototype an amortized encoder-decoder architecture.

Architecture. A DeepSets encoder [Zaheer et al., 2017] maps variable-length sparse observations $\{(k_i, \sqrt{T_i}, \sigma_i)\}$ to a fixed-size embedding via per-point MLP (64-64-64, GELU) followed by mean pooling. A conditional decoder MLP (128-64-1, GELU, Softplus output) predicts total variance $w(k, T)$ at arbitrary query points conditioned on the embedding.

Training. 300 synthetic Heston surfaces with randomized parameters at 4 dropout rates (1,200 training examples). MSE loss on dense surface values. Training takes 37 seconds (one-time cost).

Results.

The amortized model achieves $1,200\times$ speedup at competitive RMSE. The accuracy gap

Table 13: Amortized CINN vs. per-surface CINN (20 test surfaces, 40% dropout).

Method	RMSE (bps)	Inference Time	Speedup
Per-surface CINN	~ 630	$\sim 3000\text{ms}$	$1\times$
Amortized CINN	287.7 ± 58.4	2.5ms	1200$\times$

reflects generalization from 300 training surfaces; scaling to 1,000+ surfaces and adding penalty losses to the amortized training loop (transferring the same Gatheral/calendar constraints) are expected to close this gap. This demonstrates that the penalty framework is compatible with amortized inference—the constraint losses serve directly as encoder training objectives.

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